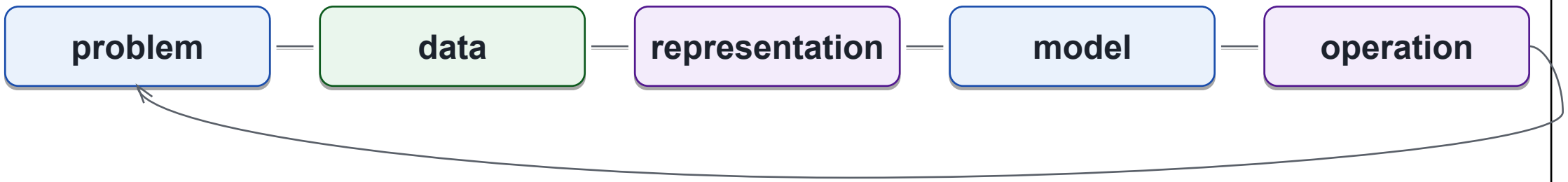
Search Data-to-Model Path

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What is a “data-to-model path”?



- MLDM **is not (**only) **about** selecting algorithms
- MLDM ***is about*** *designing, comparing and justifying paths*
- a **path** starts form a well-defined problem and its available data
- the **path** is about transforming data into validated and operational models



it is *a stronger framing —* ***it is*** *path design, not only algorithm selection*

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From CRISP-DM to a “data-to-model path” perspective

CRISP-DM
classical reference

1. business understanding
2. data understanding
3. data preparation
4. modeling
5. evaluation
6. deployment

the “data-to-model path”
adaptation

- formulate hypotheses and analytical questions
- transform data and construct representations
- build, validate and operate models
- monitor and revise the (full) path continuously

*CRISP-DM is not discarded! the **data-to-model path** adaptation makes it:*
***more** data-centric, representation-aware, multi-modal and operational*

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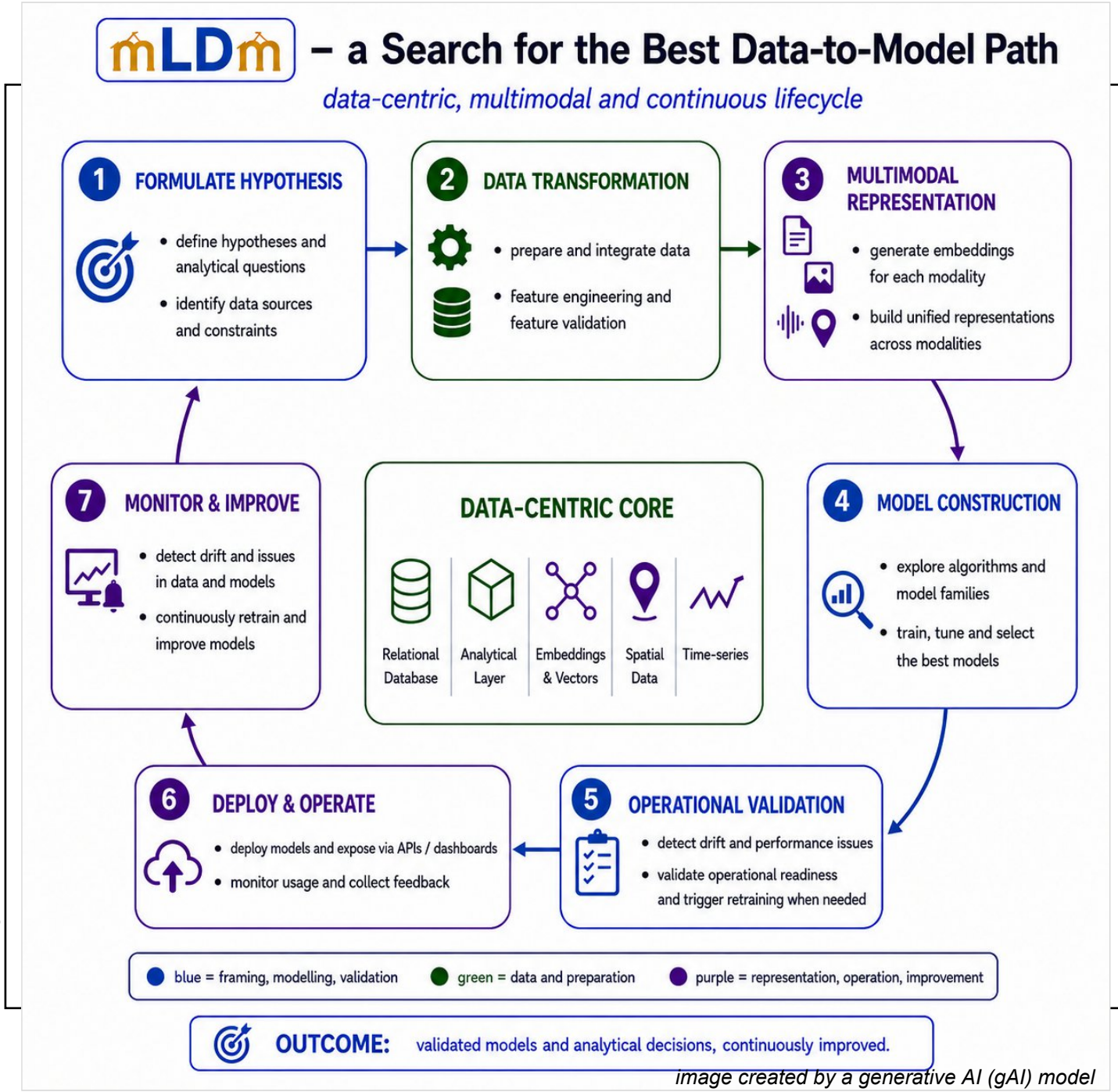


image created by a generative AI (gAI) model

MLDM
as-a-process

as-a compact **adaptation** of CRISP-DM

to-the

Best Data-to-Model Path

perspective

the process is **not a rigid checklist**

it is a **way to make explicit** the decisions, assumptions and revisions

... closer view at 1 — 2 — 3



1 — understand the problem

2 — take the information perspective

3 — go from (data) types to its representations

blue = framing, modelling, validation green = data and preparation purple = representation, operation, improvement

... closer view at 4 — 5 — 6 — 7

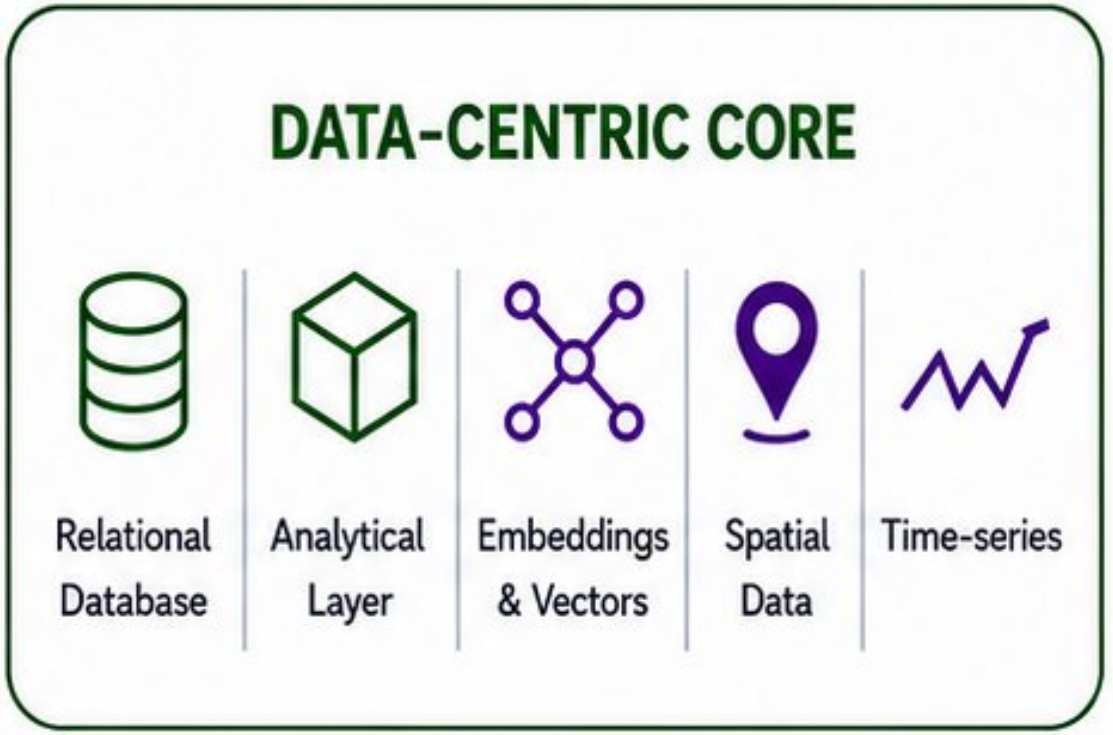


blue = framing, modelling, validation green = data and preparation purple = representation, operation, improvement

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... closer view at the “data-centric core”

data and **structure** — relational, hyper-cube, vector, geometry, time-flow, etc



OUTCOME: validated models and analytical decisions, continuously improved.

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... from algorithm-selection to “path-design”

weaker framing

1. get a dataset
2. choose a method
3. apply it to a dataset
4. report a score

stronger framing

1. define the problem and success criteria
2. construct and represent data
3. compare candidate paths
4. justify the final artifact and its limits

(some) key-notes

- a **dataset** is not just a file
- a **model** is not just an output
- a **result** is not just a score

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The data-centric perspective enables to formulate an “inverse-problem”

forward-problem

- given a well-know function: $\mathbf{F}$
- given some observed data: $\mathbf{X}$
- compute outcome: $\mathbf{y} = \mathbf{F}(\mathbf{X})$

we do **not start with true function**

we **infer** a defensible **approximation** from **observed** data

inverse-problem (*model*)

- given some observed data: $\mathbf{X}$
- given some observed outcome: $\mathbf{y}$
- infer a model approximation: $\mathbf{y} \approx \hat{\mathbf{F}}(\mathbf{X})$

inverse-problem (*structure*)

- given some observed data: $\mathbf{X}$
- ~~$\mathbf{y}$~~ i.e., there is no associated outcome
- infer X structural properties: $\widehat{\mathbf{G}}(\mathbf{X})$

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A minimal formal view

from observations with [optional target]: $\mathbf{X}$, $[\mathbf{y}]$

to learned-approximation: $\hat{\mathbf{F}}$ or $\hat{\mathbf{G}}$

supervised setting

observed-examples:	$\mathbf{X}$, $\mathbf{y}$
learned-**model**:	$\hat{\mathbf{F}}$ such that $\hat{\mathbf{F}}(\mathbf{X}) \approx \mathbf{y}$
task:	*classification*
	regression
	survival-analysis

unsupervised setting

observed-examples:	$\mathbf{X}$
learned-**structure**:	$\hat{\mathbf{G}}(\mathbf{X})$
task:	*clustering*
	association-rules
	outlier-detection

*start with the available **observations** and hypothesize about the learning **tasks** to pursue*

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A (relevant) learning setting question — “with or without a target”?

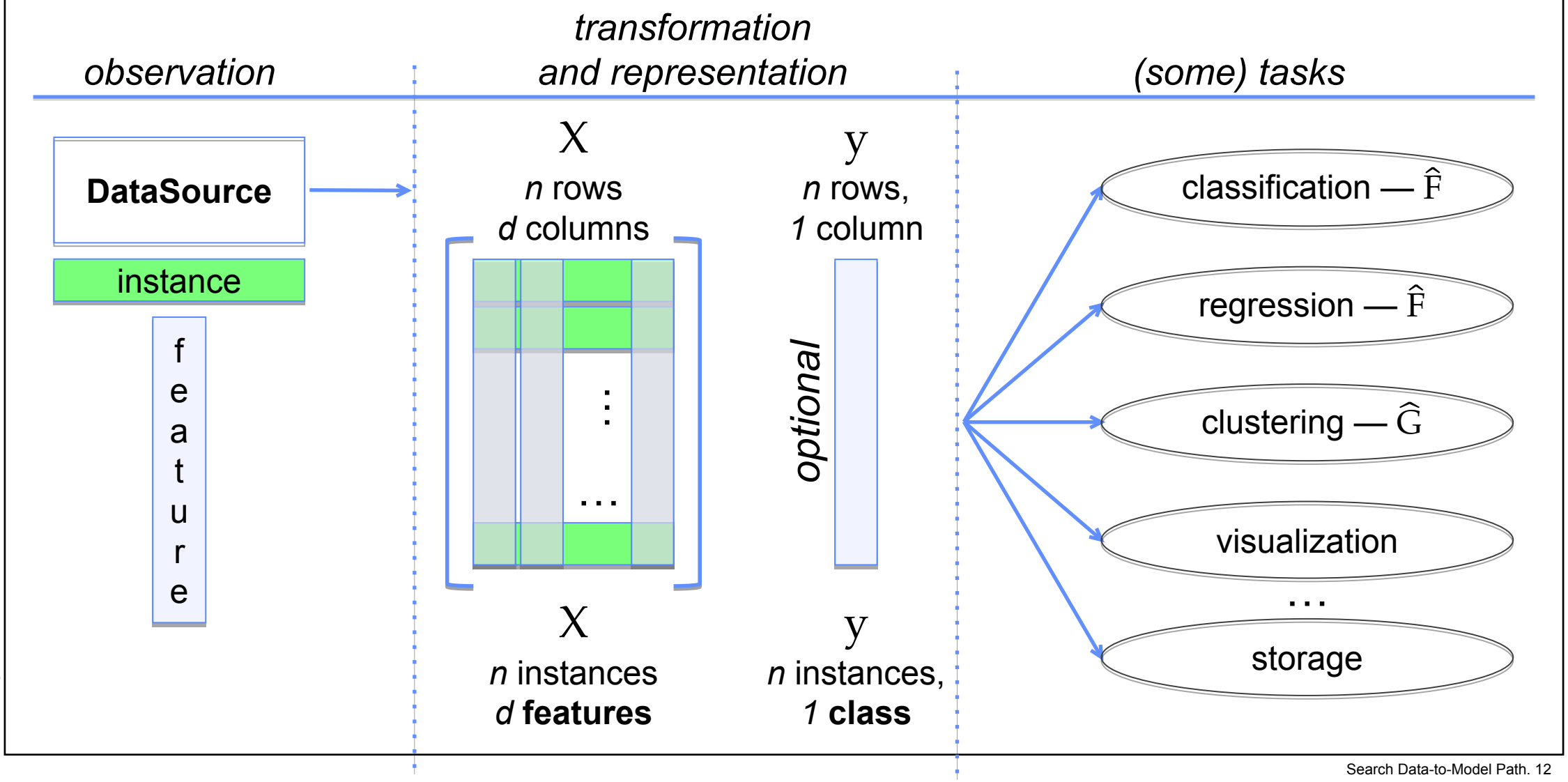
setting	observed data	goal	(usual) learning tasks
supervised learning	$\mathbf{X}$ **and** $\mathbf{y}$	learn to **predict** the value of a target (y) given the values of features (X)	classification, regression
unsupervised learning	$\mathbf{X}$ *only*	discover useful **structure** in the data	clustering, associations, anomalies

core message

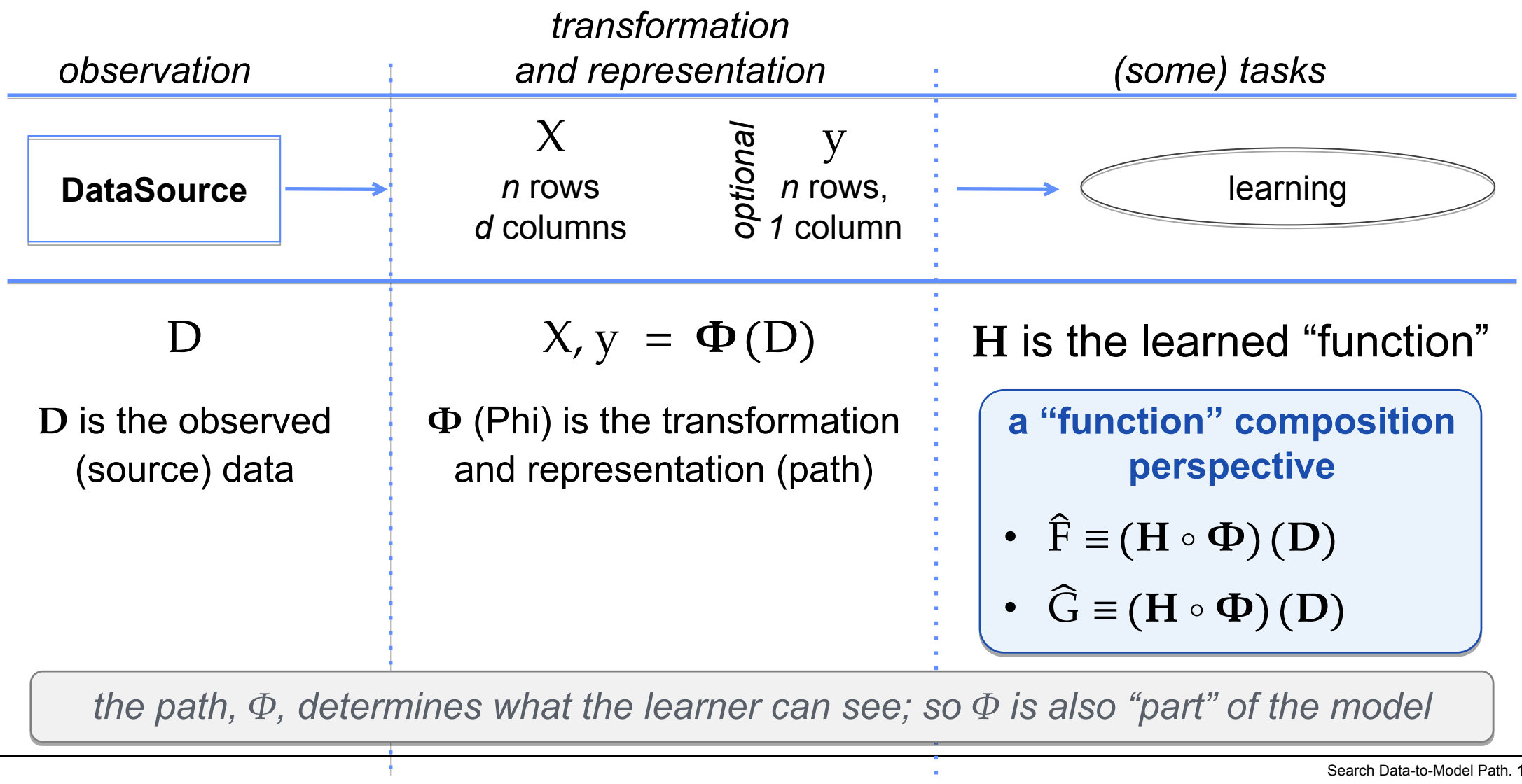
- the presence or absence of target variable, $\mathbf{y}$, changes the nature of the learning task/problem
- it also changes the way success can be evaluated

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... visual perspective of the data — X, y, and the learned "functions" — $\hat{F}$, $\hat{G}$



... and a “function” composition perspective



(some) Terminology

- data is organized as a set, $\mathbf{X}$, of instances/observations/examples/individuals
- each instance, $\mathbf{x} \in \mathbf{X}$, is a vector with d elements, i.e., $\mathbf{x} = [\, \mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_d \,]$
- an element, $\mathbf{x_i}$, of an instance, $\mathbf{x}$, is the value of a **feature** (a.k.a an **attribute**)
- each feature (attribute) represents a specific aspect relevant to the problem
- a **dataset** is composed by n instances, each with d features

Outlook	Temperature	Humidity	Windy	Play
Sunny	85	85	False	No
Sunny	80	90	True	No
Overcast	83	86	False	Yes
Rainy	70	96	False	Yes
Rainy	68	80	False	Yes
Rainy	65	70	True	No

... supervised setting — use "ground-truth" $\mathrm{F}(\mathrm{X}) = \mathrm{y}$ and learn $\hat{\mathrm{F}}(\mathrm{X}) \approx \mathrm{y}$

- the set $\mathbf{X}$ may be associated with a target/class/tag/label vector $\mathbf{y}$
- so, each, $\mathbf{x} \in \mathbf{X}$, is associated with an **a-priori known** value in $\mathbf{y}$

... i.e., there is an "oracle" with **prior-knowledge** on **how X relates to y** (the "true" $\mathbf{F}$)

- so, in this sense, supervised learning depends on a **ground-truth** (the "true" $\mathbf{F}(\mathbf{X}) = \mathbf{y}$)

binary-problem: target with binary domain, e.g., {0, 1}, {-1, 1}, {yes, no}, etc
multiclass-problem: target domain with more than two values, e.g., {0, 1, 2, ..., M-1}
oneclass-problem: target with "almost" all values being from the same class

Outlook	Temperature	Humidity	Windy	Play
Sunny	85	85	False	No
Sunny	80	90	True	No
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Rainy	68	80	False	Yes
Rainy	65	70	True	No

... unsupervised setting — use X and learn $\hat{G}(X)$

- an instance is **NOT** associated with a target value, so **NO prior-knowledge**

... i.e., the **instances** themselves, $\mathbf{X}$, represent the "structure" within the data

(in this sense) the learning goal is to **extract inherent structure** of data, $\hat{\mathbf{G}}(\mathbf{X})$

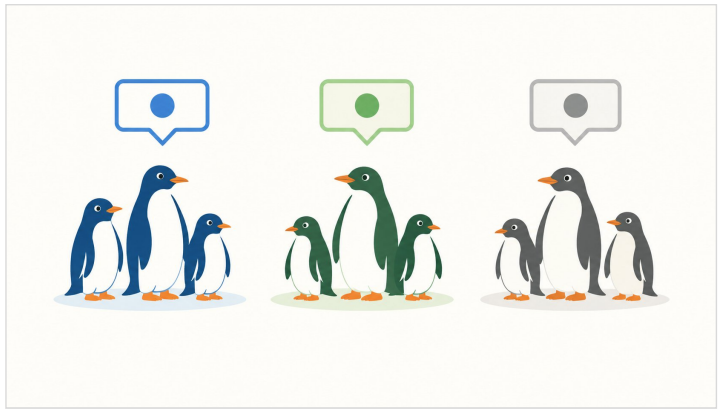
- useful in **exploratory analysis** as it automatically **identifies structure** in data

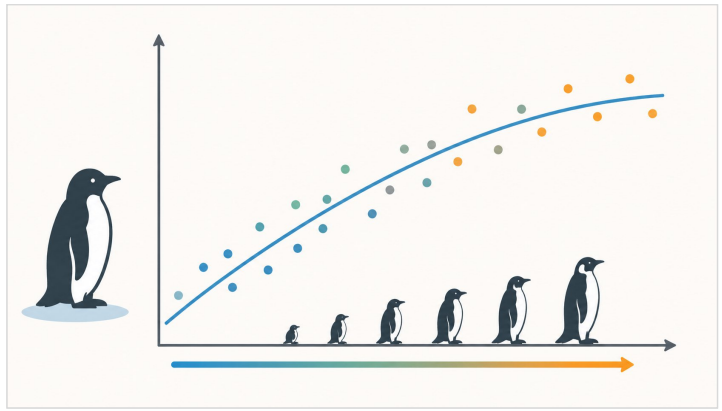
… where it is impossible or impractical for a human to propose trends in data,

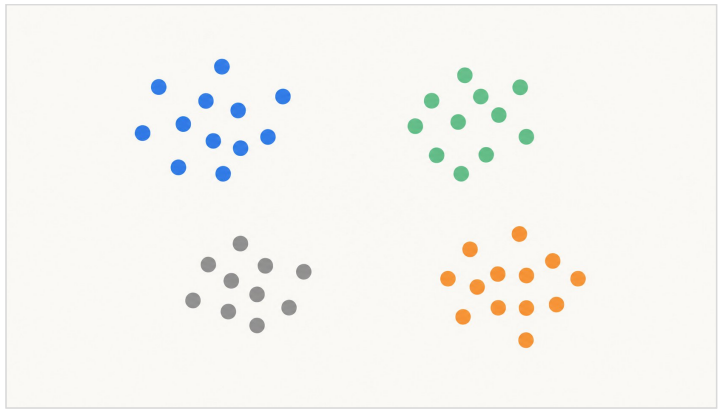
unsupervised learning can provide insights to be used **to test hypotheses**

- e.g., if an analyst is trying to segment consumers, unsupervised clustering methods would be a starting point

The learning task follows from the available observations







classification

predict a label

so, $\mathbf{y}$ **is a category**

e.g., $\mathbf{y}$ is a species-name

regression

predict a numeric value

so, $\mathbf{y}$ **is a number**

e.g., $\mathbf{y}$ is the body-mass

clustering

discover groups

so, $\mathbf{y}$ **is absent**

e.g., physical similarity

the same data can support different questions (i.e., different "problem formulations")

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... other pattern-discovery problems

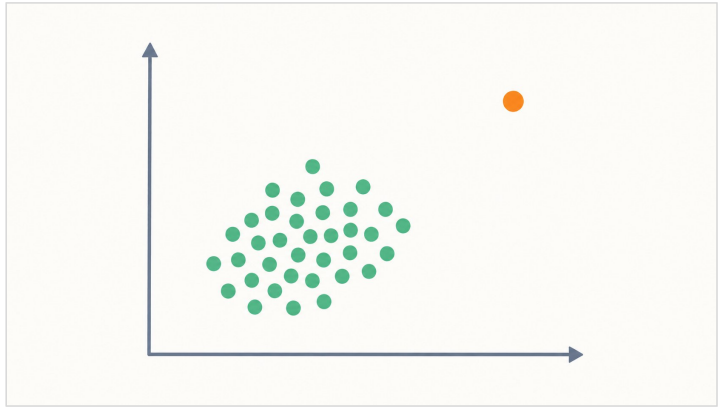


association-rules

co-occurrence patterns

so, $\mathbf{y}$ **is absent**

e.g., items often together



anomaly-detection

identify unusual cases

so, $\mathbf{y}$ **is absent**

e.g., a huge penguin

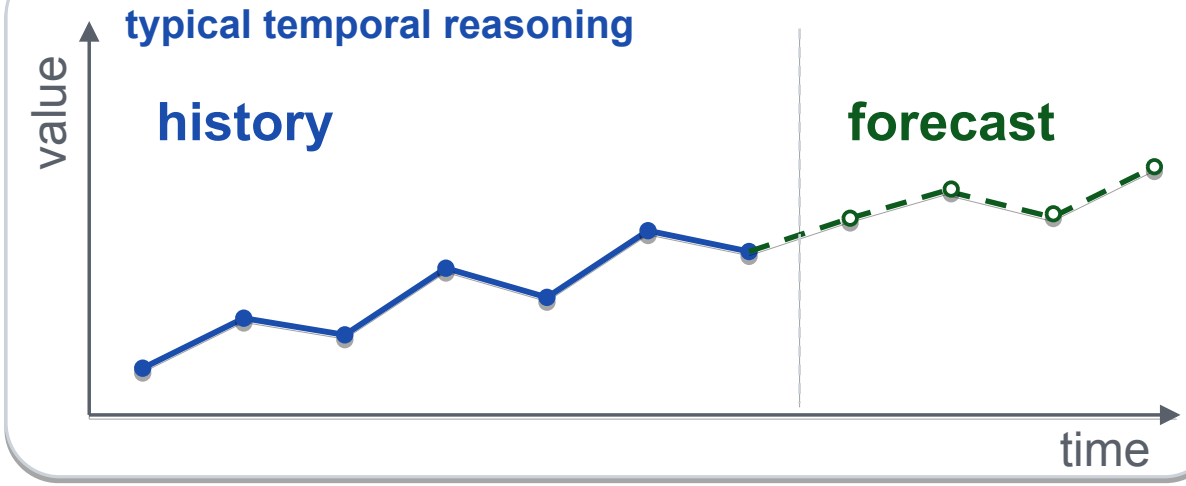
may not have a predefined target, y, but still need a clear definition of an useful structure

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When “time is part of the problem” — (some) time-series tasks

core-idea

- observations are ordered in time
- time dependency is information
- future must not leak into past



learning setting — not inherently supervised or unsupervised

- temporal structure defines the problem, learning setting depends on the task

e.g., look at a 10-second heart-rate and **classify** as “normal” or “arrhythmia”

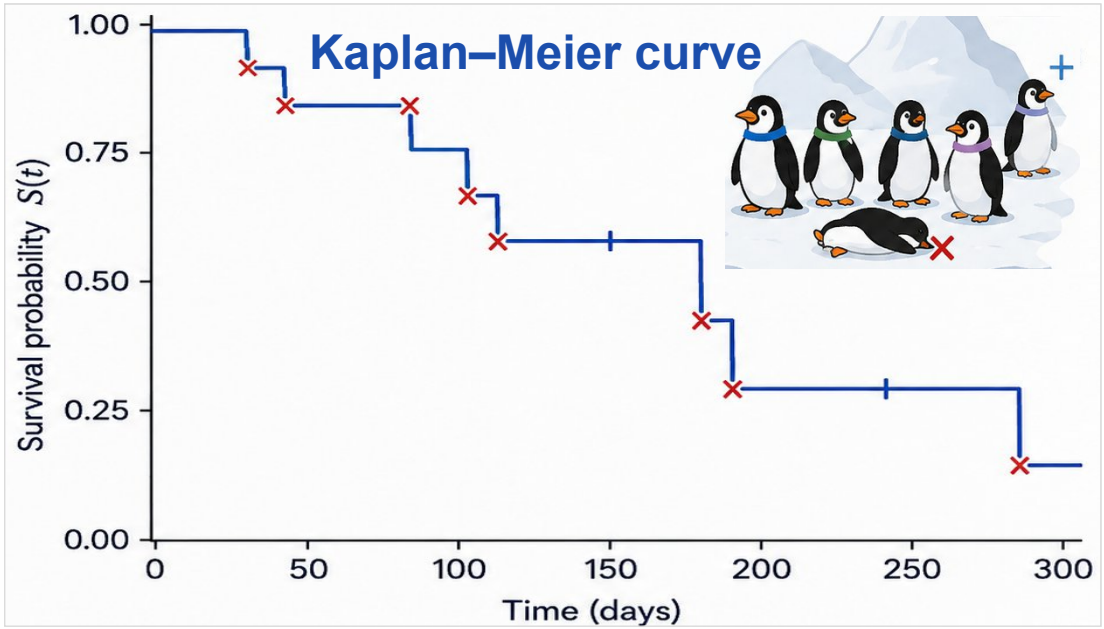
e.g., identify the data stream drops or spikes that are out of its usual **structure**

time-series analysis is defined by temporal structure — not by a single learning setting

When the goal is time-to-event (with censoring) — “survival analysis”

core-idea

- observation about time and event
- aims to predict the time-to-event
- some may exit before event



learning setting — inherently supervised

- requires ground-truth composed of both an exact duration and an event label
- uses ground-truth to extract patterns on how changes in features shifts target

censoring is the key distinction — the event time is not fully observed for every case

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What changes when the (most used) problem/task changes?

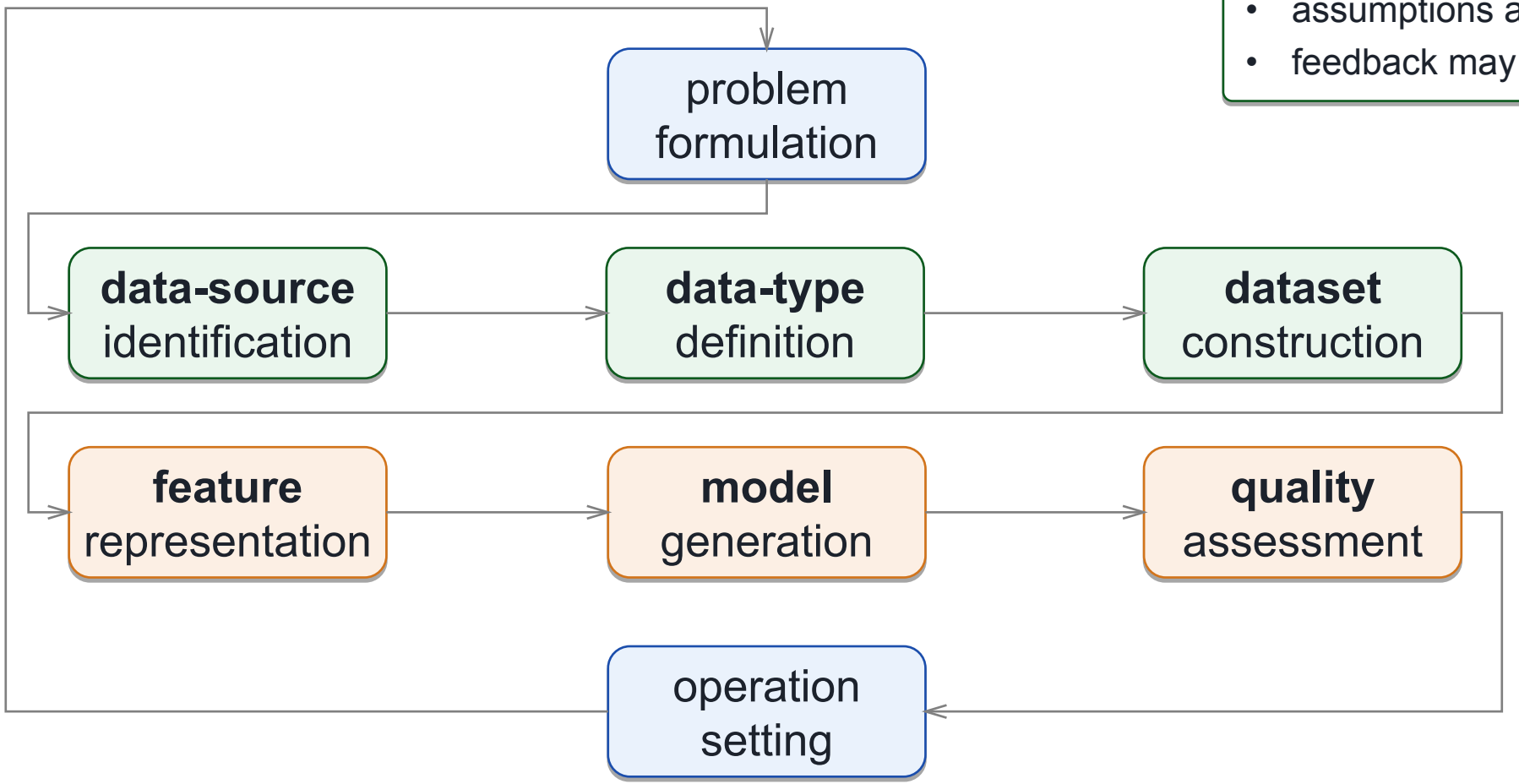
decision layer	classification	clustering	time-based
target	class-label, y	no explicit y	future value or sequence
observation unit	a labelled case	one object to group	time-indexed record
features	evidence for label	similarity representation	lags, windows, trends
success criterion	predict labels well	meaningful groups	useful forecast / pattern
risk	target leakage	poor distance geometry	future leakage

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A (possible) data-to-model path

this is not a linear checklist

- options are compared
- assumptions are revised
- feedback may trigger change



this "path view" is an operational meaning of a "search for the best data-to-model path"

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An illustrative example — “digital content engagement”

scenario

- a platform recommends videos to users
- a recommendations is shown during a session
- *goal*: predict whether the user will engage

why is it useful?

- uses several source tables
- requires a clear observation unit
- has temporal feature-availability risks
- supports several alternative paths

this example is
not about **building** a
recommender system (yet)

it **is** about **seeing how** a
data-to-model path is defined

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... illustrative example — formal problem definition

component	decision
decision-moment	a recommended video is about to be shown in a user session
analytical-question	will the user engage with the recommended video?
target	`engaged` $\in \{0, 1\}$
target-definition	1 if the user clicks or watches at least 30 seconds
observation-unit	one row per `<user, session, recommended-video>`
task	classification

the prediction must be made before observing the engagement outcome

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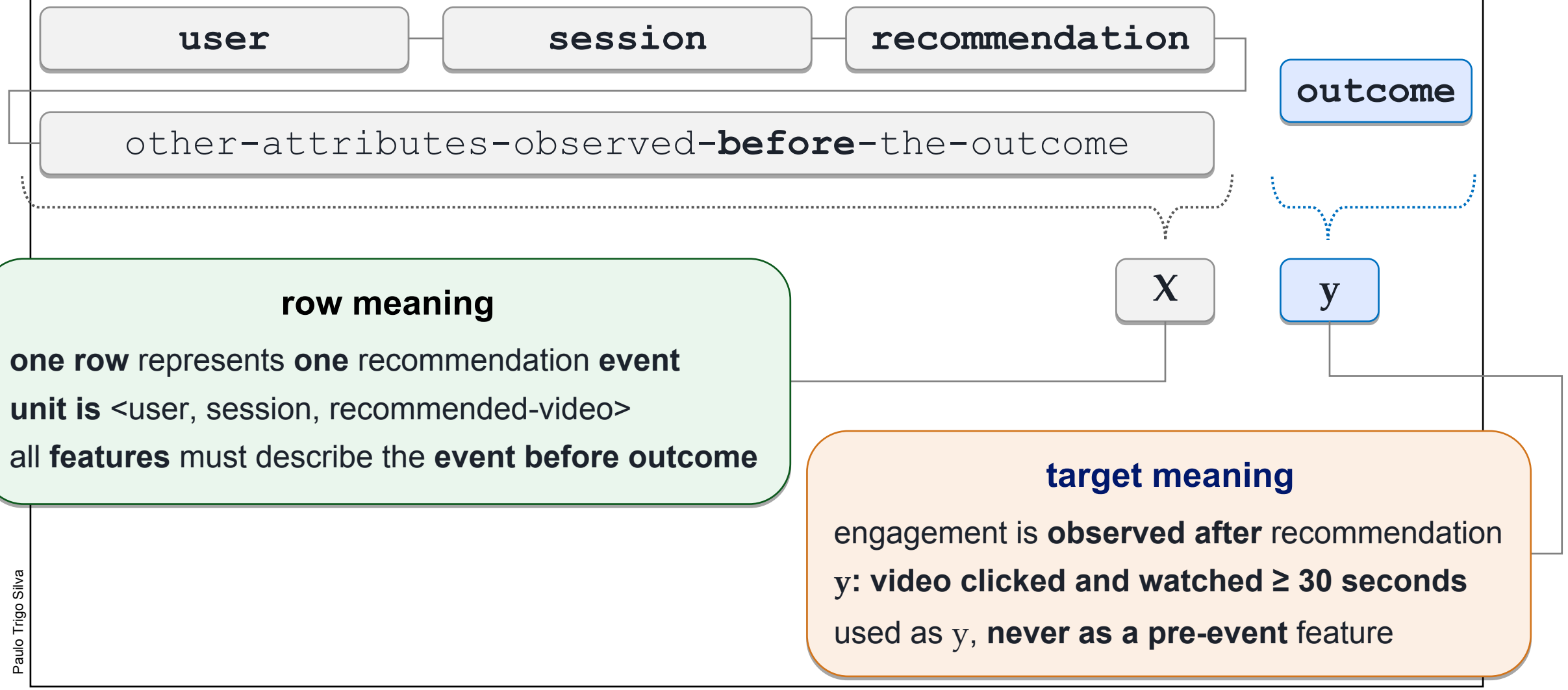
... illustrative example — source tables, attributes and roles

source table	**main-attributes**, with primary-key and *candidate-key*	**role-in-the-path**
`user`	`id, age_group, geo_region`	user profile context
`session`	`id_user, date_time, device, language`	temporal and device context
`video`	`id, title, category, duration`	content metadata
`recommendation`	`id, id_video, id_user, session_date_time, rank`	item actually recommended
`interaction`	`id_recommend, date_time, clicked, watched_seconds`	evidence used to define y

*the analytical **dataset** does **not** already **exists**; must be constructed from these sources*

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... illustrative example — construction of one row (i.e., one instance)



row meaning

one row represents **one** recommendation **event**

unit is <user, session, recommended-video>

all **features** must describe the **event before outcome**

target meaning

engagement is **observed after** recommendation

y: video clicked and watched ≥ 30 seconds

used as y, **never as a pre-event** feature

... illustrative example — quality and leakage — a path may “fail” early

availability questions

- would this feature be **known at prediction time**?
- was this **aggregate computed before** the event?
- does the **feature depend on the target**?

path-level risks

- target leakage
- future information
- invalid or inconsistent data

be aware *— a path can become invalid before any learner is trained*

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... illustrative example — candidate features

feature	type	source	availability rule
`session_hour`	temporal / circular	`session.date_time`	known before recommendation
`device`	categorical	`session.device`	known before recommendation
`video_category`	categorical	`video.category`	known before recommendation
`title_represent`	text-derived / “embedding”	`video.title`	known before recommendation
`previous_views_7d`	numeric aggregate	*past interactions*	computed only before `session.date_time`
`avg_watch_seconds_7d`	numeric aggregate	*past interactions*	must exclude current outcome

***(very) important rule** — a **feature** is **valid only if** it available at the decision moment*

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... illustrative example — features are hypothesis

H1 — time matters

represent `session_hour` cyclically

test if behaviour trend is "periodical"

H2 — content matters

use `category` and `title` representation

test if metadata improves prediction

H3 — history matters

derive `previous_views_7d`

test if recent behaviour adds information

data-type influences the **space of possible feature representations** and their **admissible transformations**

feature-design is a **kind of hypothesis formulation** that is expressed as a **representation path**

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... illustrative example — leakage checks

risk	bad construction	defensible construction
temporal leakage	`previous_views_7d` uses interactions after `session_hour`	use only interactions before session_time
target leakage	`avg_watch_seconds_7d` includes current recommended video outcome	exclude the current recommendation event
source shortcut	`id_video` is also used to encode a small set of outcomes	prefer metadata or control identifier effects
feature availability	feature exists in logs stored only after the session	keep only already available pre-decision attributes

... illustrative example — one candidate path

component	candidate-path A
problem	classification — predict `engaged` ∈ {0, 1}
dataset	one row per <user, session, recommended-video>
features	`session_hour, device, video_category, title_represent, previous_views_7d, avg_watch_seconds_7d`
representation	cyclic encoding, one-hot categories, text-derived vector, numeric aggregate
audit	all features available before recommendation aggregates exclude current event
learner	start by exploring simple and interpretable methods

this is a candidate path; this is (possibly) not the final answer

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... illustrative example — what must this path be compared with?

alternative dimension	possible variation
representation	replace title string by tf-idf or content embedding
features	remove text features or add older-history aggregates
observation unit	one row per <user, session> engagement regardless of each specific video recommendation
learner	compare simple reference with tree-based, nearest-neighbour or neural-network methods
operational assumption	test whether the same features are available in deployment

*a candidate path becomes meaningful through **comparison** with **defensible** alternatives*

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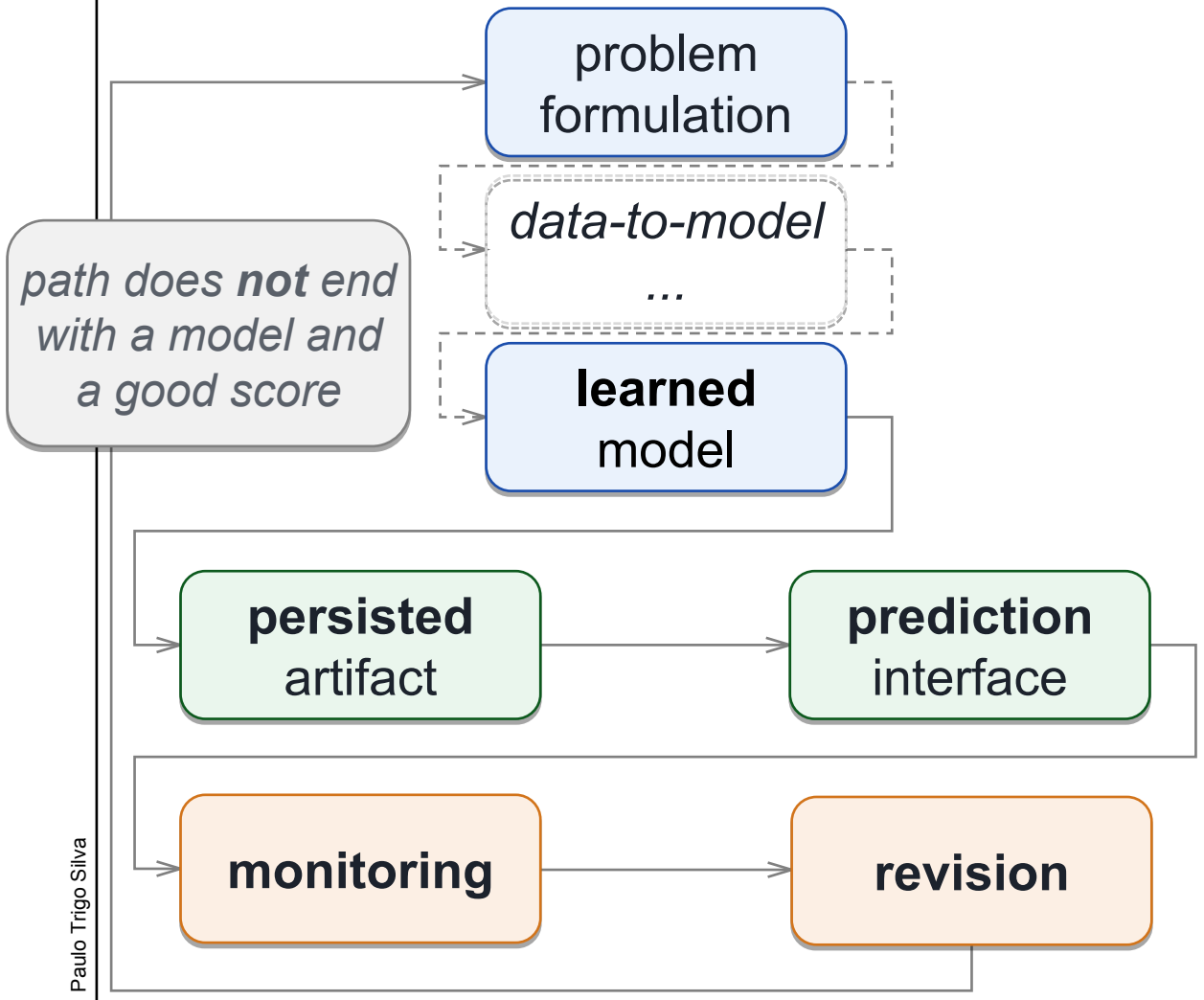
(some layers of) the data-to-model search space

layer	examples of decisions
problem	classification, regression, clustering, ranking
data	sources, observation-unit, grain, time-reference
types	numerical, categorical, temporal, spatial, text, media
representation	one-hot, cyclic-encoding, tf-idf, embeddings
features	derived ratios, aggregates, lags, interactions
quality	missingness, invalid values, leakage, availability
learner	simple reference, learner family, hyper-parameters
operation	persistence, interface, monitoring, revision



is a search over defensible decisions; it is not a single algorithmic step

The overall operational setting-and-assessment are part of the path...



persisted artifact
prediction interface

- keeps transformations with artifact
- checks inputs assumptions
- serves predictions (or decisions)

monitoring | revision

- monitor data-drift
- reassess assumptions
- revise path when needed

important-concept: **data-drift**
new data is out-of-distribution (**OOD**)
new data is out-of the training domain

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(some) main ideas



- each path is a sequence of explicit (founded) decisions
- (usually) the dataset must be constructed; not just loaded
- the problem formulation determines the search space
- data-types are derived from data-sources and “narrow” options
- data-representations determines what the learner can “see”
- each feature set express an hypothesis
- the leakage control (e.g., temporal) is part of the path
- themodel search only makes sense after the path is defensible

A final “one sentence” message



the search for the best data-to-model path is the disciplined search for defensible paths

from: **problem definition and acquired data**

to: **validated, operational and continuously re-evaluated models**

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